MIL-B-52199B(ME)
13 July 1983
SUPERSEDING
MIL-B-52199A(ME)
25 October 1968

MILITARY SPECIFICATION

BRACKET, OUTBOARD MOTOR, FOR ALUMINUM HALF-PONTON

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a welded bracket with an oak transom for attaching outboard motors to light tactical pontons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-W-21157	- Weldment, Steel, Carbon and Low Alloy (Yield Strength 30,000 - 60,000 PSI).
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWING

ME

TA13200E0991	- Outboard Motor Bracket for Light Tactical Aluminum Half-Ponton.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

ANSI/AWS D 1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 165 - Electrodeposited Coatings of Cadmium on Steel.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The bracket shall be as shown on Top Assembly TA13200E0991 and as specified herein.

3.1.1 Drawings. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawing to insure correct fit, assembly, and operation of the bracket. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subject to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

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3.3.1 Material deterioration and control. The bracket shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Treatment and painting. The bracket metal shall be cleaned, treated, and painted in accordance with MIL-T-704, type B. The clamp-screw assembly shall be cadmium plated in accordance with ASTM A 165, minimum thickness 1 mil.

3.5 Identification marking. The bracket shall be identified in accordance with MIL-STD-130.

3.6 Workmanship. All parts, components, and assemblies of the bracket, including welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, flux, and other harmful extraneous material. External surfaces shall be free from burns, sharp edges, and corners except where sharp edges and corners are required.

3.6.1 Steel fabrication. Steel used in the fabrication of the bracket shall provide original quality surface, and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

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3.6.2 Welders and welding.

3.6.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D 1.1, Structural Welding Code, Section 5, Qualification.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME and ANSI/AWS codes.

3.6.2.2 Welding. Weldments shall be in accordance with MIL-W-21157. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts are stressed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection. The first article bracket shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Samples for examination shall be selected in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The bracket shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Bracket not true to dimensions and lines, contains twists, bends, or other imperfections.
106. Treatment and painting not as specified. Presence of blisters, thin spots, lumps, scratches, or runs.
107. Identification marking missing, incomplete or illegible.
108. Workmanship not as specified.
109. Welders and welding not as specified.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. The acceptable quality level (AQL) shall be 2.5 percent defective.

No.	Defect	A	B	Comm.
110.	Preservation not as specified.	5.1.2		5.1.3
111.	Clamp screws not turned all the way in as specified.	5.2.1	5.2.2	5.2.3
112.	Four brackets not packed in each level A box as required.	5.2.1		
113.	Level A box not as specified	5.2.1		
114.	Brackets not cushioned, anchored, braced, or blocked in box as specified for level A.	5.2.1		
115.	Brackets not protected from metal to metal contact as specified.	5.2.1	5.2.2	
116.	Box closure for level A not as specified.	5.2.1		
117.	Two brackets not bundled for level B as specified.		5.2.2	
118.	Strapping not as specified.	5.2.1	5.2.2	
119.	Marking not as specified.	5.3.1	5.3.1	5.3.2

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.2 Level A. Unpainted metal surfaces of the brackets requiring the application of a contract preservative in accordance with the requirements of MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

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5.1.3 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Four (4) brackets, with their clamp screws turned all the way in and preserved as specified in 5.1.2, shall be packed in a snug fitting box conforming to PPP-B-601, overseas type, style optional. The cushioning, anchoring, bracing and blocking of the brackets within the box shall conform to the requirements of MIL-STD-1186. Pads or padding shall be secured in positions on the brackets, as required, to prevent metal to metal contact. Box assembly, closure and strapping shall be in accordance with the box specification except that the strapping shall be flat and zinc coated.

5.2.2 Level B. Two (2) brackets, with their clamp screws turned all the way in and preserved as specified in 5.1.2, shall be banded together to form a tight bundle. Each bundle shall be secured with not less than two flat steel straps, size 3/4 inch x 0.015 inch, conforming to QQ-S-781, class 1, type I, finish A. Pads or padding shall be secured in positions on the brackets, as required, to prevent metal to metal contact.

5.2.3 Commercial. The brackets, with their clamp screws turned all the way in and preserved as specified in 5.1.3, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Levels A and B. In addition to any special or other identification marking required by the contract or order (see 6.2), each cradle shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Brackets packed as specified in 5.2.3 shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The bracket is intended for use as an outboard motor support utilized with the raft section of the light tactical floating bridge.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. Degree of preservation and packing required (see 5.1 and 5.2).

d. Any special marking requirements (see 5.3.1).

6.3 First article. When a first article inspection is required, the item shall be examined and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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